

Review and Comparison of COVID-19 Prediction Studies

Mubashira Qari

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Review of Articles

Article 1: *A novel approach to predict COVID-19 using support vector machine (SVM)*

Link: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8137961/>

This study proposes a machine-learning-based diagnostic system that applies a **Support Vector Machine (SVM)** to predict COVID-19 infection severity. Using a small dataset of 200 records containing clinical attributes such as temperature, breathing rate, heart disease, and cough type, the model classified cases as *not infected*, *mildly infected*, or *severely infected*.

Through a linear kernel SVM and a 70/30 train-test split, the model achieved an **accuracy of 87%** with strong recall for the *severely infected* class, indicating its ability to minimise false negatives in critical cases. The study emphasises that machine learning can supplement slow, resource-intensive RT-PCR testing by offering rapid, data-driven assessments based on routine medical indicators. However, it also acknowledges limitations such as the small sample size and reliance on structured data rather than large-scale clinical records.

Article 2: *Decision Tree Ensembles to Predict Coronavirus Disease 2019 Infection: A Comparative Study*

Link: <https://www.hindawi.com/journals/complexity/2021/5550344/>

This paper compares multiple **decision tree ensemble algorithms**—including Random Forest, XGBoost, Bagging, AdaBoost, and specialised imbalanced-data models like **RUSBagging** and **Balanced Random Forest (RUS)**—to classify COVID-19 cases using laboratory test data.

The dataset (600 cases from São Paulo) was **highly imbalanced** (520 negative vs. 80 positive). Using **10-fold cross-validation**, results showed that ensemble methods designed for imbalance significantly improved performance:

- **RUSBagging:** AUROC = 0.881

- **Balanced Random Forest:** AUPRC = 0.561

Adding **age** as a predictor further enhanced results, reflecting its known clinical relevance. The study highlights that **metric selection (AUPRC over accuracy)** and **data-specific modelling choices** are critical for reliable medical predictions. It also recommends integrating lab tests with imaging data and expanding research to SVMs and deep learning frameworks.

Comparison and Insights

Both studies share a common goal—**using machine learning to improve early COVID-19 diagnosis**—but differ in scope and methodological rigour:

Aspect	SVM Study (NCBI)	Decision Tree Ensemble Study (Hindawi)
Algorithm Type	Linear SVM (single model)	Multiple ensemble trees (Random Forest, XGBoost, RUS variants)
Dataset Size	200 samples	600 samples
Data Type	Clinical symptoms & comorbidities	Laboratory test results
Handling of Imbalance	Not addressed explicitly	Directly addressed via SMOTE, RUS, and hybrid ensembles
Performance Metric	Accuracy & recall	AUROC, AUPRC, F1-score
Best Model Result	87% accuracy	AUROC = 0.881 (RUSBagging), AUPRC = 0.561 (Balanced RF)

Key Insights

1. Model complexity vs. interpretability:

The SVM model provides a quick, interpretable decision boundary suitable for small, structured datasets, while the ensemble trees handle larger, more heterogeneous data more robustly.

2. Handling imbalanced data:

The second paper's focus on imbalance correction and AUPRC as a fairness metric makes its results more credible for real-world medical use, where false negatives have serious consequences.

3. Feature influence:

Both studies demonstrate that incorporating clinical or demographic context (e.g., *age*, *comorbidities*) enhances predictive power.

4. Scalability and generalisation:

The ensemble study offers a stronger framework for future pandemic modelling, integrating oversampling/undersampling strategies and demonstrating adaptability to new datasets.

Area of Expertise/Interest

My professional and academic focus lies in **Data Science and Predictive Analytics**, particularly the application of **machine learning models for health and risk prediction**. The insights from these studies align closely with my interest in **data preprocessing, model tuning, and evaluation using ROC/AUC metrics**.

- The SVM model illustrates how linear classifiers can act as strong baselines for binary and multi-class health classification tasks when data volume is limited.
- The Decision Tree Ensemble research demonstrates advanced ensemble learning strategies (Bagging, Boosting, Random Forest) that I frequently apply in my own analytical work—such as evaluating model bias, optimizing hyper-parameters (`tuneLength`, `max_depth`, `n_estimators`), and validating through cross-validation.

Results Related to Area of Expertise

From a data-science perspective: - The studies reinforce the importance of **data balancing** (RUS, SMOTE) before training models—a process I apply in predictive modelling tasks like incident severity or bank marketing classification.

- Both demonstrate how **AUC-based evaluation** provides a more reliable measure of model performance than raw accuracy, especially for skewed datasets.

- The findings encourage using **hybrid or ensemble frameworks** that combine interpretability and predictive strength, aligning with my goal to develop **explainable AI (XAI)** models in healthcare analytics.

Summary

Together, these articles illustrate a transition from single-model approaches (SVM) to advanced **ensemble and imbalance-aware frameworks** (RUSBoost, XGBoost) that achieve higher predictive reliability for COVID-19 detection.

For a data scientist, the research underscores the critical balance between **algorithm selection, data characteristics, and metric interpretation**, ultimately guiding more ethical and effective use of AI in health diagnostics.
